



Grasshopper Ventures-Organic Recycling

Carbon dioxide removal prepurchase application

2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Grasshopper Ventures (Organic Recycling)

Company or organization location (we welcome applicants from anywhere in the world)

Honolulu, Hawaii. Our company is currently incubated at the University of Hawaii (Manoa) Natural Energy Institute with distributed research & engineering collaboration across USC (Cal), UC Riverside, University of Miami (FL), San Diego State University & Oak Ridge National Lab.

Name(s) of primary point(s) of contact for this application

Brandon Arns

Brief company or organization description <20 words

Modular Anaerobic Digestion System for Integrated Carbon Dioxide Removal & Energy Recovery

1. Public summary of proposed project¹ to Frontier

- a. Description of the CDR approach: Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise.

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

At its core, this project is not a technological novelty—it is a systems-level deployment architecture engineered for real-world, scalable climate impact. The individual technologies—anaerobic digestion, SOFCs, CO₂ sorbents, biochar kilns, and geopolymer mineralization—are technically validated and widely studied, many already operating at TRLs of 7–9. Our innovation lies in the strategic reconfiguration of these components into a modular, rapidly deployable, and operationally integrated carbon removal platform.

We are fundamentally addressing the *deployment bottleneck* in CDR—how to reach gigaton scale in a distributed, durable, and economically viable way. Our solution is a containerized, plug-and-play infrastructure that co-locates with the emission source: the organic waste stream itself. Rather than centralizing infrastructure and transporting biomass, we bring the system to the point of waste generation—university campuses, QSRs like Chipotle, warehouse retailers like Costco—unlocking massive, untapped, and fragmented biomass flows that current BECCS and CCS systems cannot economically access.

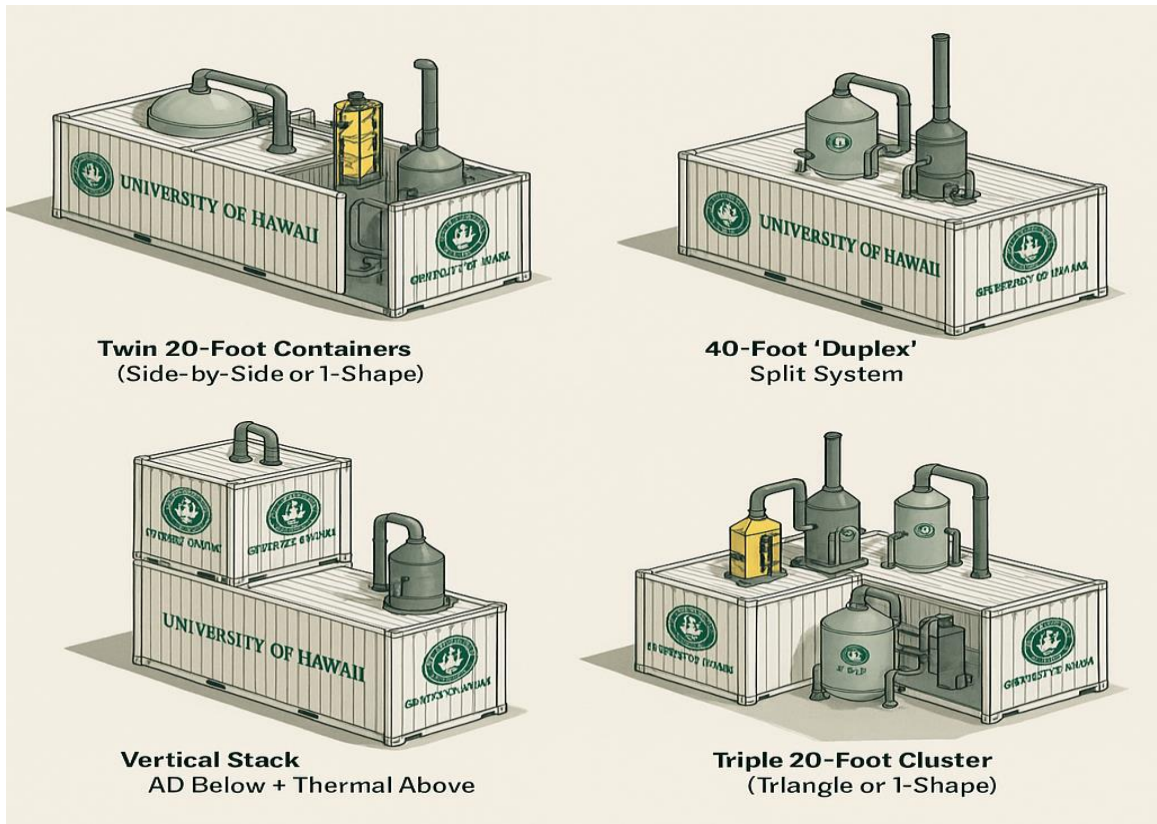
The system is engineered to address the specific concerns outlined in Frontier’s RFP regarding methane leakage, digestate emissions, and centralized biogas upgrading, while enabling scalability through modular deployment. The system begins with the collection of distributed organic waste—including pre- and post-consumer food scraps—from sources such as universities, grocery retailers, and food service chains. This feedstock, which would otherwise emit methane or CO₂ via landfilling or composting, is fed into a high-efficiency digester that generates biogas and solid digestate. Critically, the system avoids co-digestion of residues like crop waste that would have decomposed aerobically, minimizing “intentionally-produced” methane.

Biogas from the digester is directed through a LEAF (Living Emission Abolish Filter) unit, where halophilic methanotrophs neutralize any fugitive methane emissions. The cleaned methane then powers an onsite SOFC, which generates low-emission electricity while producing a concentrated CO₂ exhaust stream. This CO₂ is captured using Ni-doped CaTiO₃/CaO solid sorbents with 90–95% efficiency. The sorbents are regenerable and can either be reused or mineralized into concrete products.

This system directly addresses Frontier’s concerns about traditional AD pathways. Methane emissions are eliminated through fully sealed systems and biofilters; digestate emissions are avoided by converting solids to biochar rather than applying them to land; and rather than upgrading biogas to RNG and venting CO₂, we use SOFCs for onsite power and immediate carbon capture. The system also produces valuable co-products—electricity, biochar, and carbon-negative concrete—avoiding the limitations of RNG-only strategies and delivering measurable climate and economic value.

Each modular system is designed to process ~100–400 tons of organic waste per year, depending on site constraints. For modeling, we are using a standardized throughput to ensure conservative, comparable CO₂ removal projections (~149 tCO₂ net/year). Higher throughput sites (e.g., universities, stadiums) can host multi-unit clusters or run the system at higher daily loads to increase carbon removal at lower marginal cost. All four designs (below) can be built within a similar capital expenditure (CAPEX) range, but with slight variations depending on layout complexity, equipment redundancy, and site-specific integration. The real difference is not in what they include, but how it’s spatially arranged. Each site installs a modular AD system that processes food and organic waste, generating biogas for onsite electricity and converting digestate into biochar and carbon-sequestering concrete—ensuring as much of the carbon in the digestate is captured as possible. What makes our model transformative is its centralized MRV aggregation and our Corporate-Wide AD Deployment Strategy: individual deployments feed into a single carbon project per brand (e.g., “Costco Carbon Loop” or “Chipotle Carbon Project”), dramatically reducing MRV complexity and credit issuance overhead. This

architecture enables coherent, tradable, large-scale carbon removal portfolios for otherwise decentralized emitters. Ex scenario: 2,900+ Chipotle locations estimated net CDR: ~20,000–40,000 tons CO₂/year. Looking ahead, we are leading the development of a new University/College “Race to Carbon-Negative Campus” coalition, inspired by the global Race to Zero. This initiative will invite universities to host modular carbon removal systems, while embedding CDR into campus sustainability plans, climate curricula, and student research.



From a financial architecture standpoint, we’re leveraging performance-based leasing and climate infrastructure capital, allowing host sites to subscribe to a CDR-as-a-Service model without upfront investment. This not only derisks adoption but ensures alignment between CDR performance and revenue streams. Carbon credits are monetized centrally, while co-products—electricity, avoided tipping fees, carbon-negative construction materials—enhance the host site’s bottom line and sustainability profile.

Strategically, this model turns each modular unit into both a carbon sink and a public research node, integrated with university labs for MRV and learning, and aligned with corporate ESG strategies. In effect, this is not simply an engineering deployment—it is a scalable blueprint for distributed, high-verifiability, durable CDR infrastructure, designed to bring gigaton-scale removal within reach by decentralizing deployment and centralizing verification.

Our platform is uniquely positioned as a best-in-class solution for modular BECCS. It is easily deployable via standard 20- or 40-foot containers, rapidly installable, and designed to integrate directly with existing built environments like campuses, apartment communities and commercial properties. Higher throughput sites (e.g., universities, stadiums) can host multi-unit clusters or run the system at higher daily loads to increase carbon removal at lower marginal cost. This avoids the land-use conflicts

and hauling emissions typical of centralized BECCS or DAC facilities. Host sites also benefit economically through avoiding hauling and tipping fees (\$8,000–\$15,000/year), electricity generation, and sustainability visibility. Planned pilots are being designed where the system doubles as a research and public engagement platform within campus living labs. At national scale, deploying ~500,000 units could remove over 75 MtCO₂/year from the U.S. food waste stream alone, while enabling localized economic benefits and circular carbon pathways.

- b. Project objectives: What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

We are developing a modular, on-site carbon removal (CDR) system that transforms unavoidable food and organic waste into permanent carbon sinks using anaerobic digestion, biochar pyrolysis, and CO₂ mineralization into cement. (Planned Visual UHawaii Frontier Project)



The system operates in a closed-loop configuration optimized for durability and verifiability. Organic waste is collected onsite from university campuses, food courts, and grocery outlets using source-separated collection. This feedstock undergoes anaerobic digestion (AD), producing biogas (a mixture of CH₄ and CO₂) and solid digestate. To prevent methane leakage—a key concern in traditional AD systems—we integrate a Living Emission Abolish Filter (LEAF) that uses halophilic methanotrophs to eliminate fugitive emissions. The resulting methane is combusted in a Solid Oxide Fuel Cell (SOFC), generating clean electricity and a high-purity CO₂ exhaust stream. This CO₂ is captured with ~90–95% efficiency using Ni-doped CaTiO₃/CaO solid sorbents, which can be regenerated or mineralized directly. Simultaneously, the solid digestate is pyrolyzed at 500–650°C to produce biochar, a highly stable carbon form suitable for either soil amendment or construction-grade cement. Finally, the captured CO₂ and biochar are co-integrated into geopolymer cement formulations, locking carbon into durable, carbonate-bound building materials.

Our near-term scale-up plan targets modular installations at university campuses, distributed QSR locations, and food logistics hubs. Each unit is capable of scaling up or down depending on waste volume per location per year. Globally, food waste exceeds 900 million tons per year. If just 20% of that is converted using our system, it could yield 0.6–1.2 GtCO₂/year in durable CDR, making our

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

approach climate-relevant at global scale.

Embedded economics significantly improve cost-effectiveness: host sites avoid \$8,000–\$15,000 annually in tipping and hauling fees; SOFC-generated electricity offsets utility costs or can be sold at \$5–\$10/tCO₂; biochar can be monetized or offset soil purchases at \$10–\$30/tCO₂; and cement applications displace traditional clinker while sequestering carbon at ~\$10–\$15/tCO₂.

Our MRV system is robust and integrated. Onboard flow meters measure biogas volumes and CO₂ concentrations in real time. Biochar batches are tested for carbon content and H:C ratio to determine permanence, and all injected cement is tracked via CO₂ tags linked to production batches. Our pilot sites will include MRV stations operated by students under university supervision, and we align our protocols with emerging standards from Isometric and peer-reviewed biochar quantification methods.

This project fully aligns with Frontier’s innovation focus on anaerobic digestion and modular BECCS. Our approach directly addresses the key weaknesses of conventional AD systems—including methane leaks, digestate emissions, and low additionality. Rather than upgrading methane to RNG, we eliminate leaks through LEAF biofiltration and extract electricity via SOFCs while capturing the associated CO₂. Our system is co-located with waste generators, drastically reducing transportation emissions and land use. Moreover, it produces multiple co-benefits: clean energy, local soil enhancements, climate-labeled construction materials, avoided tipping fees, and powerful public education platforms through university living labs.

Ultimately, we are delivering a replicable and economically viable platform for distributed carbon removal at scale. Our model supports gigaton-scale deployment through modular design, embedded incentives, and rigorous verification—paving the way toward climate-aligned, community-powered infrastructure that captures and stores carbon where it is most efficiently and permanently locked away.

- c. Risks: What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Our modular carbon removal system—combining anaerobic digestion (AD), Solid Oxide Fuel Cell (SOFC) electricity generation, and CO₂ capture through biochar and mineralized concrete—presents a range of technical, execution, MRV, ecosystem, financial, and policy risks. To responsibly scale this distributed platform, we’ve developed a comprehensive mitigation strategy tailored to each risk category. On the technical side, integrating digestion, fuel cells, sorbent capture, and pyrolysis in a compact unit introduces system complexity. We mitigate this by standardizing our modules, using field-tested interoperable controls, and relying on commercially available high-TRL components. SOFC reliability is addressed by pre-treating biogas, using ruggedized biogas-rated fuel cells, and ensuring vendor support and diagnostics. To ensure consistent biochar quality, we monitor pyrolysis temperatures in real-time and perform batch-level H:C ratio testing per EBC and Isometric standards to validate permanence.

From a project execution standpoint, delays in permitting, siting, or interconnection—particularly in commercial or urban environments—could slow deployment. To avoid this, we begin with host sites like universities and food service operations on private land, where approvals are faster. We’re also developing a “Permitting Playbook” and engaging regulators early. Supply chain risks such as delays in container retrofitting or sourcing sorbents are mitigated through multiple regional vendors, buffer inventory, and co-development with university and industry partners. To address scaling challenges, we are creating a technician training pipeline with university partners and using plug-and-play system architecture with remote diagnostics.

For MRV (measurement, reporting, and verification), the primary risks are inaccurate carbon

accounting and potential double-counting of credits. Our units are equipped with real-time sensors and blockchain-linked tracking for biogas flow, digestate output, and biochar production. Each carbon removal ton is assigned a unique ID and independently verified (e.g., via Isometric). To eliminate credit overlap, all credits are contractually assigned to Frontier or designated CDR buyers, and host sites (e.g., universities) are restricted from claiming emissions reductions related to the project.

Ecosystem and community risks include potential odor, noise, or visual impacts that could trigger local opposition. We mitigate this by enclosing all digestion and pyrolysis systems, incorporating odor filtration and sound insulation, and designing the units to be campus- and community-friendly with interpretive signage and public dashboards. In terms of waste quality, post-consumer inputs may be contaminated with plastic or chemicals. We address this with robust pre-sorting, mechanical screening, and prioritization of pre-consumer food waste, alongside routine testing of biochar batches before use in soil or cement.

Financial risks include the potential for high CapEx during initial FOAK (First-Of-A-Kind) deployments and uncertain carbon credit pricing in voluntary markets. Our FOAK units are currently optimized at \$150K–\$200K, with modeled NOAK reductions of over 40% due to manufacturing efficiencies and economies of scale. Co-product revenue streams—including electricity generation, avoided waste hauling costs, and biochar/cement sales—further reduce effective carbon removal costs to below \$100/ton. To address volatility in carbon pricing, we are pursuing a diversified revenue strategy combining carbon credits, waste offsets, and ESG-aligned corporate buyers, with Frontier’s prepurchase serving as a critical validation milestone. Maintenance risk is mitigated by designing for modular upgrades, predictive diagnostics, and service packages supported by trained technical staff.

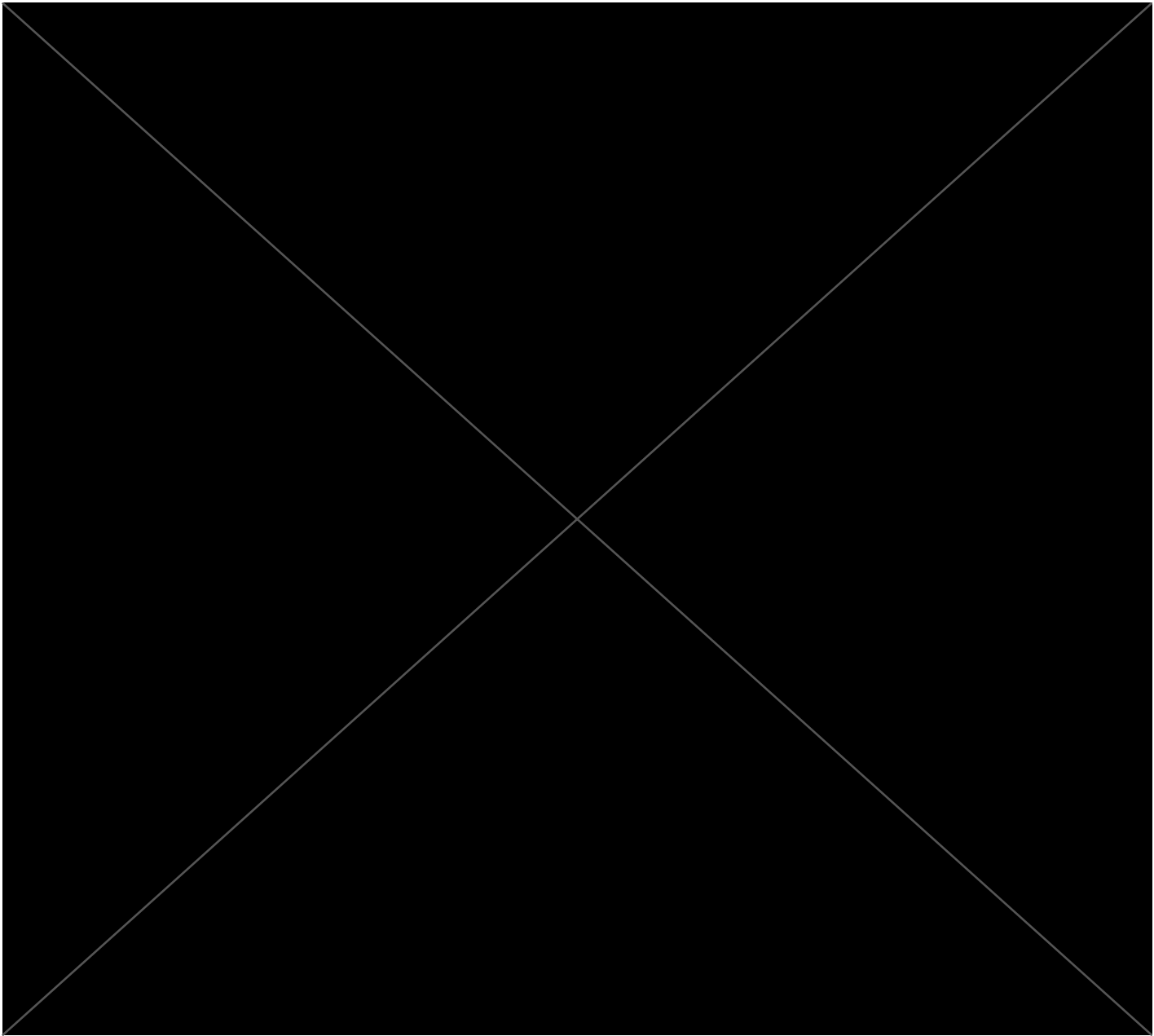
Policy and reputational risks are also front of mind. Regulatory uncertainty—such as future changes to 45Q tax credit eligibility or MRV protocols—could affect the business case. However, our model does not rely on subsidies, and we engage actively with policy stakeholders and carbon standards bodies (e.g., ISO, IPCC). To ensure lasting community trust, we adopt transparency practices based on Project Vesta’s framework and Arnstein’s Ladder of Participation. Each host site incorporates community engagement features, such as signage, learning modules, and accessible data reporting.

Our project has been engineered for multi-dimensional resilience—from hardware to MRV, from policy navigation to public trust. Through robust technical design, distributed deployment, and stakeholder-centered execution, we turn risk into opportunity and build a system that can scale responsibly, transparently, and affordably. With Frontier’s catalytic support, we are confident in delivering a best-in-class, low-cost, durable carbon removal platform at the gigaton scale.

- d. Proposed offer to Frontier: Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	149 Co2 Tons (net)
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Q1 2026-Q1 2027
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$399

Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$595
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³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).